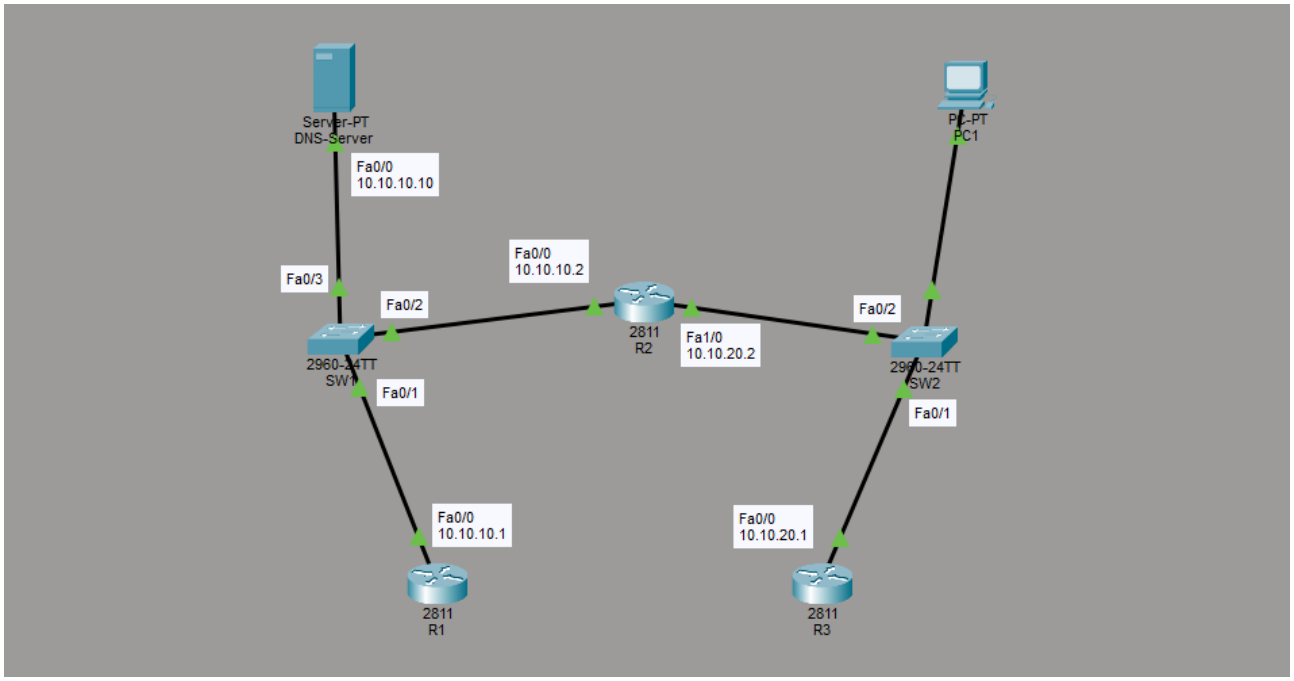
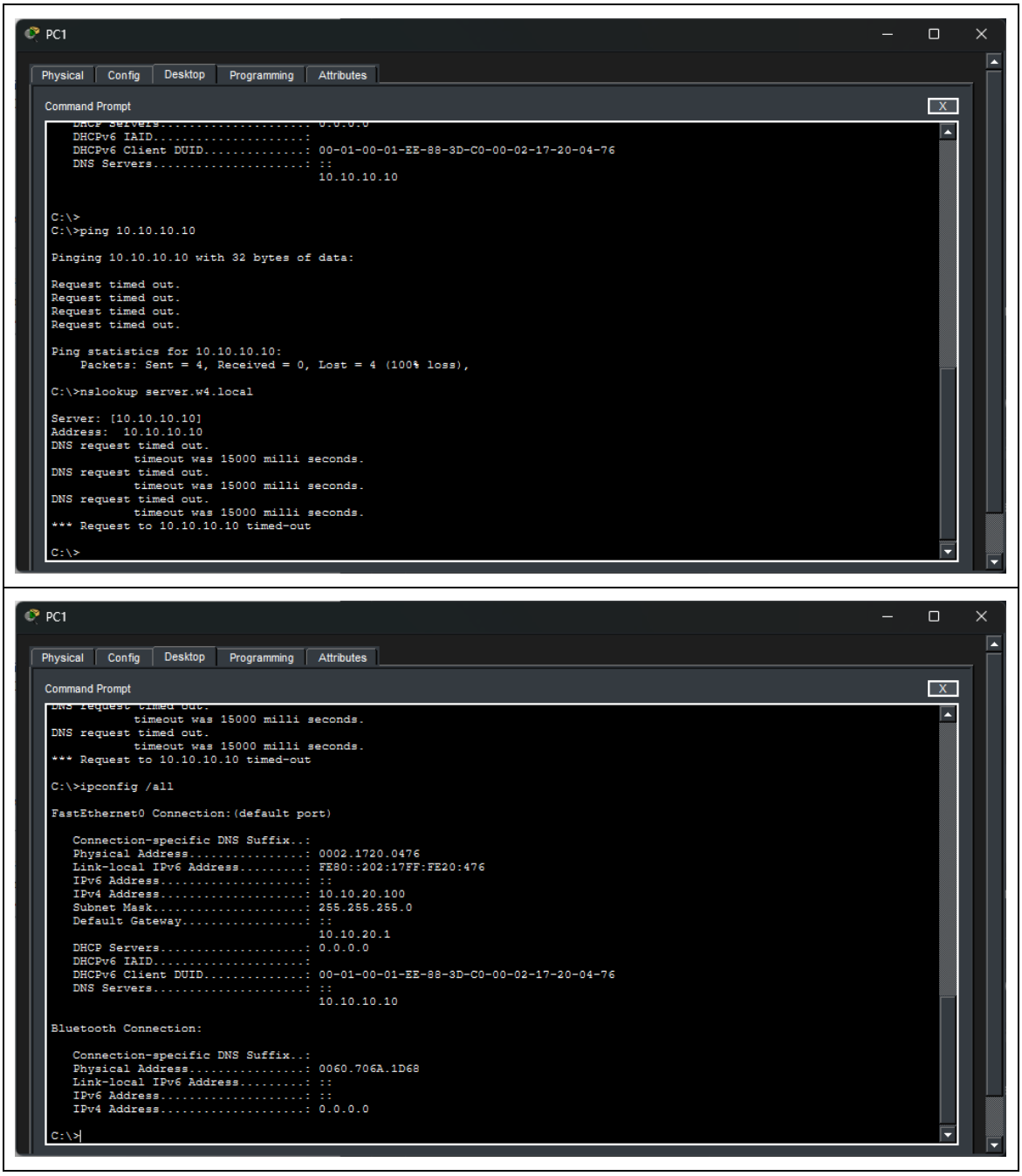
NET512 — Weekly Practical

NAME	Christiaan van Wyk	WEEK	Week 4
STUDENT NO.	20261193	DATE	25 September 2026

Topology

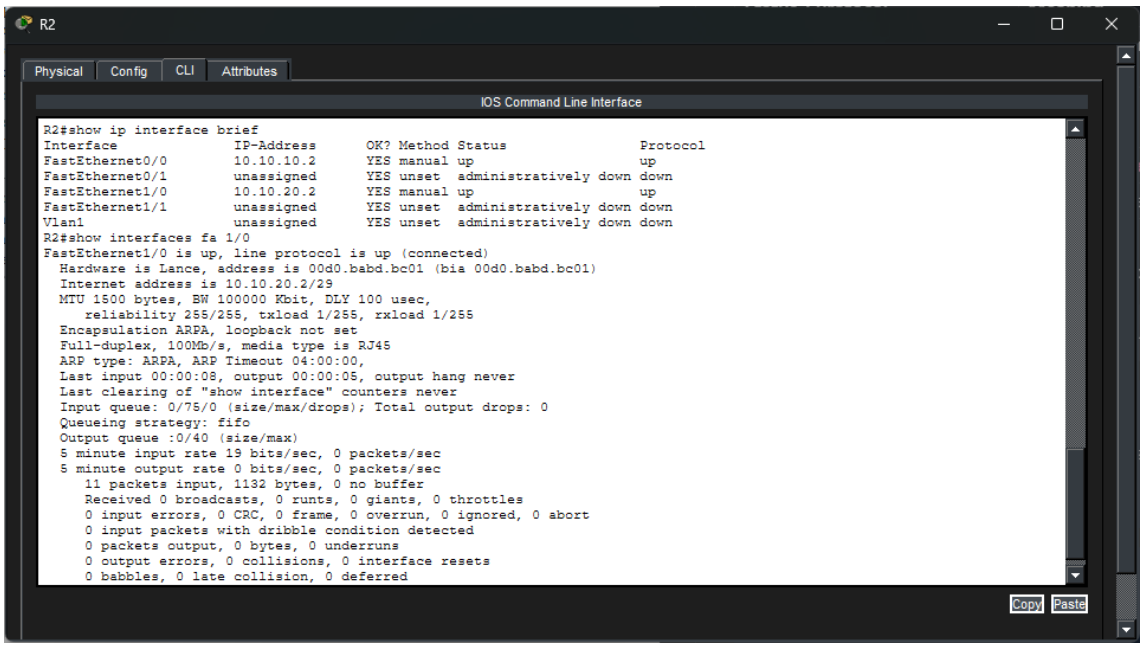


Task 22 — Confirm failure



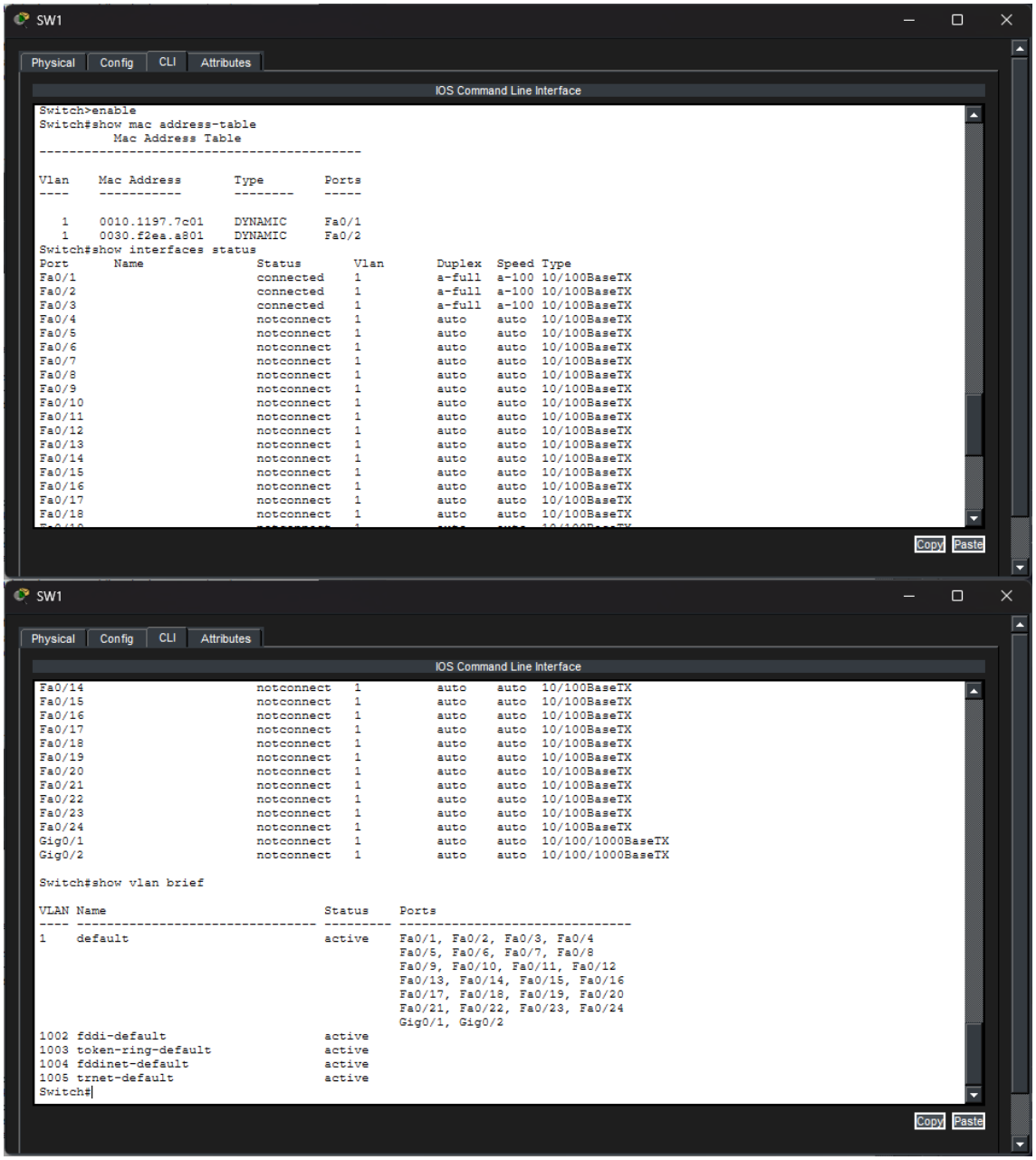
Task 23 — Work through OSI layers

L1 - Physical



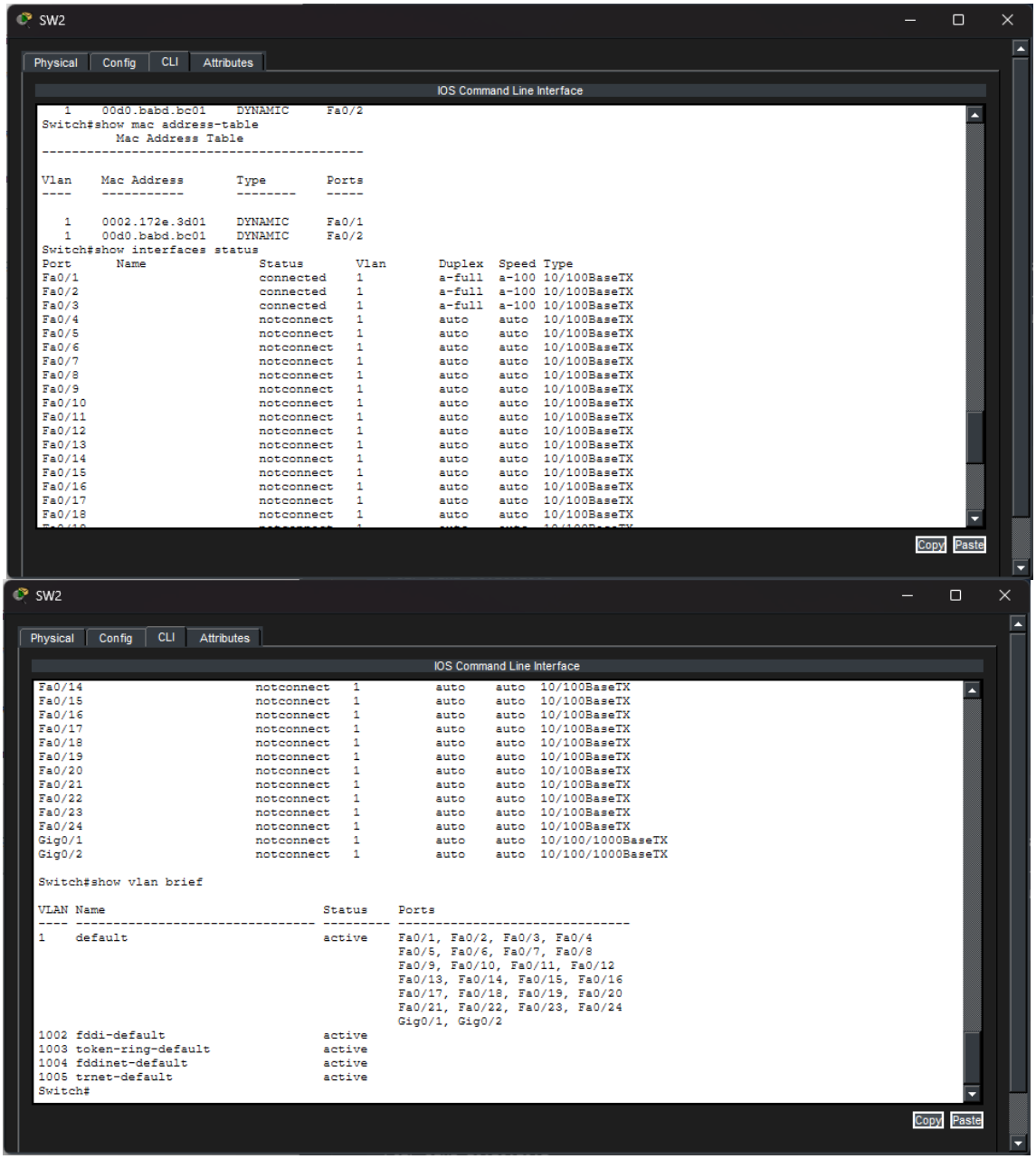
L1 - Pass

SW1


SW1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>enable
Switch#show mac address-table
Mac Address Table
Vlan Mac Address Type Ports
1 0010.1197.7c01 DYNAMIC Fa0/1
1 0030.f2ea.a801 DYNAMIC Fa0/2
Switch#show interfaces status
Port Name Status Vlan Duplex Speed Type
Fa0/1 connected 1 a-full a-100 10/100BaseTX
Fa0/2 connected 1 a-full a-100 10/100BaseTX
Fa0/3 connected 1 a-full a-100 10/100BaseTX
Fa0/4 notconnect 1 auto auto 10/100BaseTX
Fa0/5 notconnect 1 auto auto 10/100BaseTX
Fa0/6 notconnect 1 auto auto 10/100BaseTX
Fa0/7 notconnect 1 auto auto 10/100BaseTX
Fa0/8 notconnect 1 auto auto 10/100BaseTX
Fa0/9 notconnect 1 auto auto 10/100BaseTX
Fa0/10 notconnect 1 auto auto 10/100BaseTX
Fa0/11 notconnect 1 auto auto 10/100BaseTX
Fa0/12 notconnect 1 auto auto 10/100BaseTX
Fa0/13 notconnect 1 auto auto 10/100BaseTX
Fa0/14 notconnect 1 auto auto 10/100BaseTX
Fa0/15 notconnect 1 auto auto 10/100BaseTX
Fa0/16 notconnect 1 auto auto 10/100BaseTX
Fa0/17 notconnect 1 auto auto 10/100BaseTX
Fa0/18 notconnect 1 auto auto 10/100BaseTX
Copy
Paste
SW1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Fa0/14 notconnect 1 auto auto 10/100BaseTX
Fa0/15 notconnect 1 auto auto 10/100BaseTX
Fa0/16 notconnect 1 auto auto 10/100BaseTX
Fa0/17 notconnect 1 auto auto 10/100BaseTX
Fa0/18 notconnect 1 auto auto 10/100BaseTX
Fa0/19 notconnect 1 auto auto 10/100BaseTX
Fa0/20 notconnect 1 auto auto 10/100BaseTX
Fa0/21 notconnect 1 auto auto 10/100BaseTX
Fa0/22 notconnect 1 auto auto 10/100BaseTX
Fa0/23 notconnect 1 auto auto 10/100BaseTX
Fa0/24 notconnect 1 auto auto 10/100BaseTX
Gig0/1 notconnect 1 auto auto 10/100/1000BaseTX
Gig0/2 notconnect 1 auto auto 10/100/1000BaseTX
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/6, Fa0/7, Fa0/8
Fa0/9, Fa0/10, Fa0/11, Fa0/12
Fa0/13, Fa0/14, Fa0/15, Fa0/16
Fa0/17, Fa0/18, Fa0/19, Fa0/20
Fa0/21, Fa0/22, Fa0/23, Fa0/24
Gig0/1, Gig0/2
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Copy
Paste

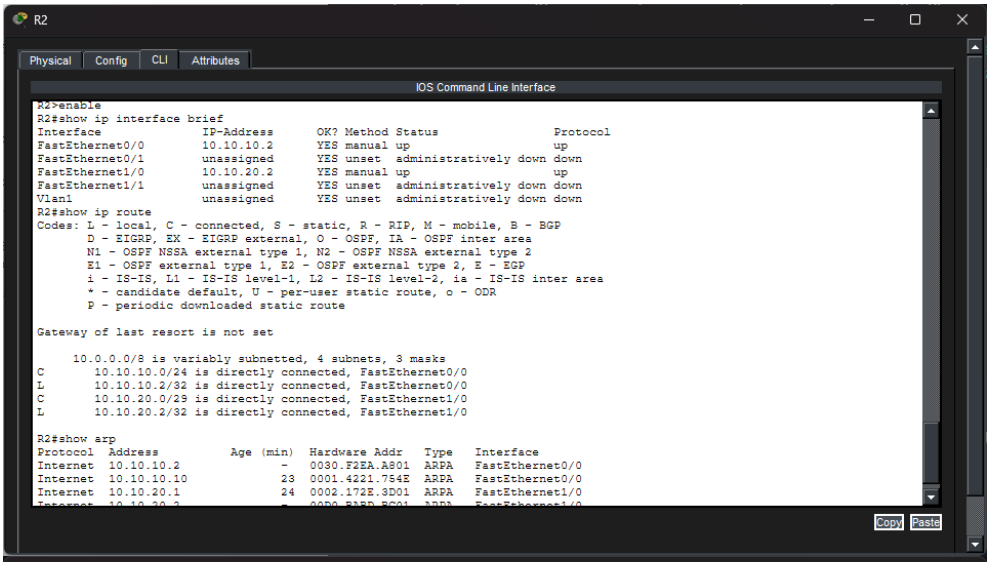
SW1 - Pass

SW2



SW2 - Pass

L2 – Pass



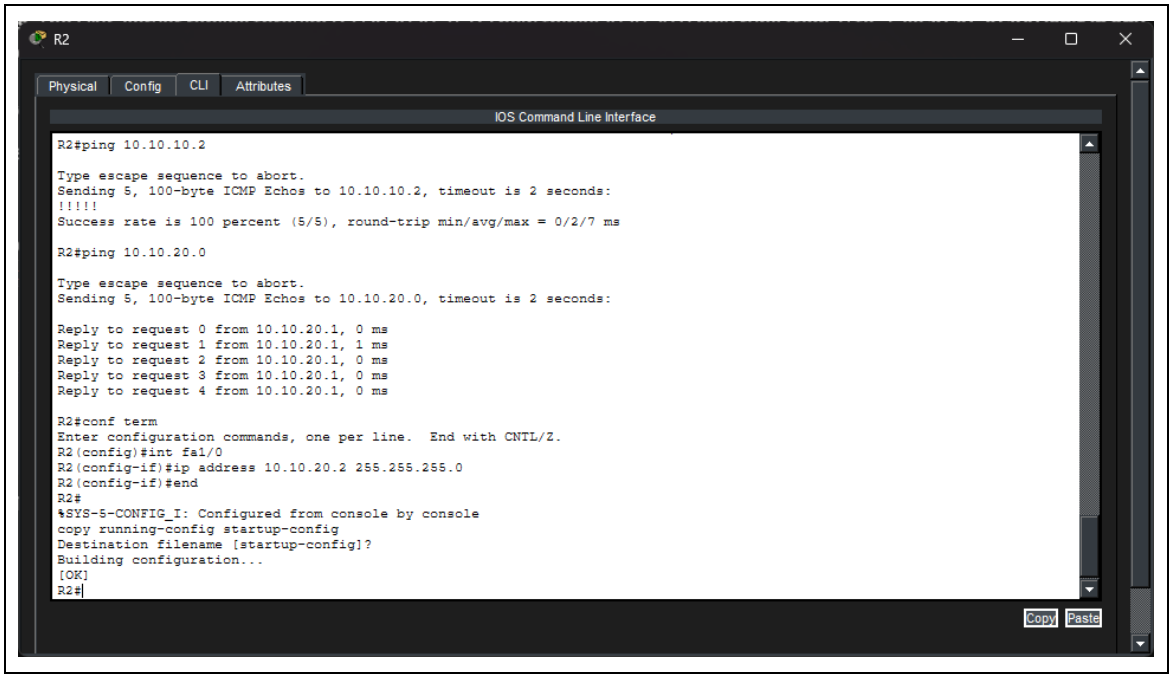
L3 - Fail

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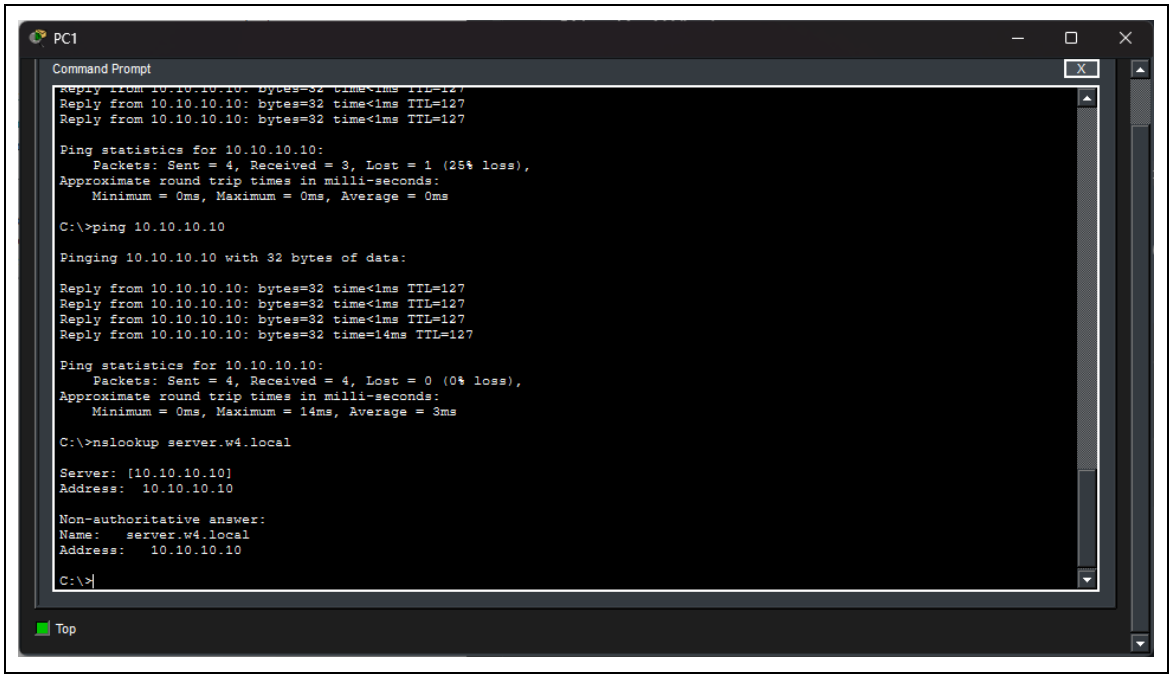
Task 24 – Identify cause

The fault lies in Layer 3 – Network. The issue is that the FA1/0 interface on R2 has a misconfigured subnet mask. Its mask is set to 255.255.255.248 (/29) compared to everything else in the network that uses a /24 mask. This means that while R2 itself can reach the DNS server, everything that needs to go through its FA1/0 interface, while the traffic can reach the server, the return path fails as the mask fails anything above .7 (PC1 is .100) thus failing the ping.

Task 25 — Apply fix



Task 26 — Re-verification



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Reflection Questions

Task 27 – The Fault

The fault was a misconfigured mask for one of the interfaces on R2. The interface had a mask of 255.255.255.248. This meant that while PC1 could reach the server (the other R2 interface was fine) anything that the server tried to send back failed because with that subnet mask the FA1/0 interface is unable to see anything with an address over .7 (PC1 is .100).

The fault was discovered by going through the layers of the OSI model one by one. Layer 1 passed scrutiny as all interfaces on R2 showed up/up and the connections between all devices in the network showed green.
Layer 2 passed as all MAC addresses of the interfaces were in order visible in the table and all interfaces were on the correct vlans.
The fault was discovered in Layer 3 when looking at the interface brief and seeing that interface FA1/0 had a prefix of 29 compared to FA0/0 which had a prefix of /24. This means that one of them had to be misconfigured, and after testing pings in from R2 and PC1 it was found to be FA1/0 (with /29). As that prefix means devices in our network would not be visible with their current IP address.

The fix was simple, just re-assigning the address for the interface with the correct mask (255.255.255.0).